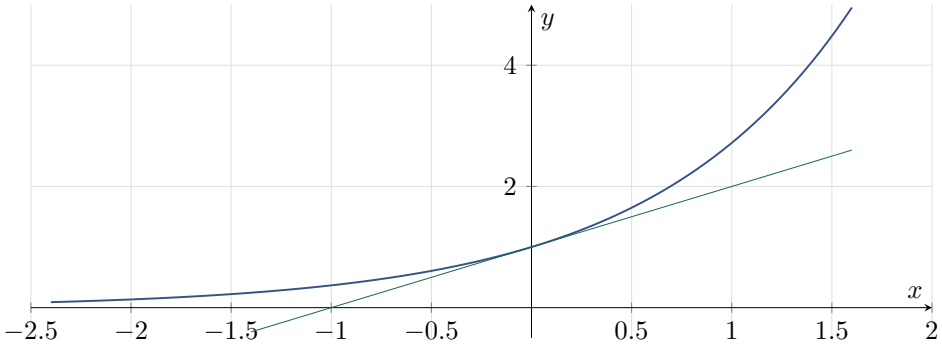
9.1 Defining a new function

Definition and results

The exponential function is the unique function f differentiable on $\mathbb{R}$ satisfying $f' = f$ and $f(0) = 1$. Its existence and uniqueness are accepted. It is written exp, then e^x with $e = \exp(1) \approx 2.71828$. The number e is a constant; x is the variable. The exponential is strictly positive.

Status and justification. Existence and uniqueness accepted in accordance with the programme. Positivity is deduced, after the algebraic properties accepted below, from $\exp(x) = \exp(\frac{x}{2})^2$ and from $\exp(x)\exp(-x) = 1$.

Worked example. $\exp(0) = 1$, $\exp(1) = e$, $\exp'(0) = 1$. The tangent at 0 is therefore $y = 1 + x$. The notation e^x extends the properties of powers to every real exponent.



e^x (blue) and its tangent $1 + x$ at 0 (green).

Watch out

$e^0 = 1$, not 0. The function never takes the value 0, even for negative x.

Immediate application. What is the equation of the tangent to e^x at 1?

9.2 Algebraic properties and comparisons

Definition and results

For all real x,y, $e^{x+y} = e^x e^y$, $e^{-x} = \dfrac{1}{e^x}$, $e^{x-y} = \dfrac{e^x}{e^y}$. For integer n, $(e^x)^n = e^{nx}$. The function is strictly increasing on $\mathbb{R}$, so $e^u = e^v \iff u = v$ and $e^u < e^v \iff u < v$.

Status and justification. The addition relation is accepted in the common pathway; it can be explored further via uniqueness. Positivity and $\exp' = \exp > 0$ give strict increase. This justifies the equivalences for equations and inequalities.

Worked example. $e^{2x}e^{1-x} = e^{x+1}$. $e^{3x-2} = e^{x+4}$ is equivalent to $3x - 2 = x + 4$, so x=3. $e^{2x} \leq 1 = e^0$ is equivalent to $x \leq 0$.

Watch out

e^{x+y} is a product, in general never $e^x + e^y$. We do not solve $e^x = 2$ with a logarithm, which is not studied in Première.

Immediate application. Simplify $\dfrac{e^{3x}}{e^{x-1}}$; solve $e^{x+2} > e^{2x}$.

9.3 Differentiating and studying

Definition and results

For real a, $t \mapsto e^{at}$ is differentiable and its derivative is ae^{at}. For $e^{at+b} = e^b e^{at}$, the derivative is ae^{at+b}. The product and quotient rules apply with the derivatives already known. The sign of the exponential often simplifies the study of the sign of the derivative.

Status and justification. The formula for differentiating e^{at} is accepted; the case with b follows by taking out the constant factor e^b. The general chain rule is not needed here.

Worked example. For $f(x) = (x + 1)e^{-x}$, $f'(x) = e^{-x} - (x + 1)e^{-x} = -xe^{-x}$. Since $e^{-x} > 0$, the derivative has the sign of $-x$: global maximum 1 at 0.

Watch out

Do not forget the factor a or the negative sign in e^{-x}. The exponential factor is positive, not necessarily increasing.

Immediate application. Differentiate $g(t) = 3e^{-2t}$ and study its direction of variation.

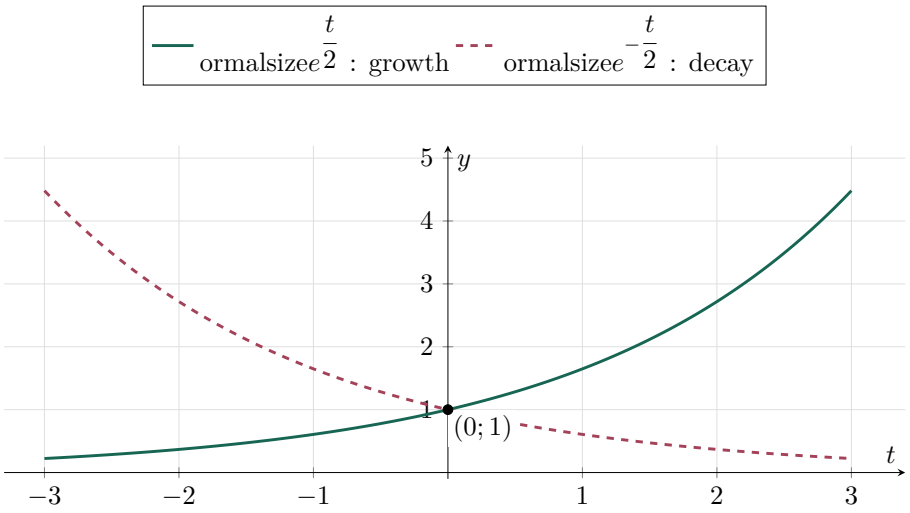
9.4 From discrete to continuous

Definition and results

A model $Q(t) = Q_0 e^{kt}$, with $Q_0 > 0$, represents growth if $k > 0$, decay if $k < 0$, constancy if $k = 0$. For a fixed step h, $\dfrac{Q(t+h)}{Q(t)} = e^{kh}$. The values at integer times form a geometric sequence with common ratio e^k. The unit of k is the inverse of the unit of time.

Status and justification. The ratio follows from $e^{k(t+h)} = e^{kt}e^{kh}$. The direction of variation is deduced from $Q\prime(t) = kQ_0e^{kt}$ and the positivity of the remaining factor.

Worked example. $Q(t) = 100e^{-0.2t}$, t in hours: $Q(0) = 100$, $\dfrac{Q(t+1)}{Q(t)} = e^{-0.2}$. A simulation can look for the first whole number of hours at which Q falls below 50; the exact equation is not solved with a logarithm in this chapter.



The curves for $k = \dfrac{1}{2}$ and $k = -\dfrac{1}{2}$ are symmetric about the vertical axis. The line styles also distinguish them without colour.

Watch out

The relative rate per discrete unit of time is $e^k - 1$, not exactly k; they are close only for small k.

Immediate application. For $Q(t) = 5e^{0.3t}$, express $Q(n)$ and $\dfrac{Q(t+2)}{Q(t)}$.

9.5 Building numerically and proving an inequality

Definition and results

To approximate the solution of $y' = y$, $y(0) = 1$ with a step h, Euler replaces the increment by hy: $y_{k+1} = (1 + h)y_k$. On $[0; 1]$ with $h = \frac{1}{n}$, the approximation of e is $\left(1 + \frac{1}{n}\right)^n$. This construction is experimental. The inequality $e^x \geq 1 + x$ is obtained by studying a function.

Status and justification. For $g(x) = e^x - 1 - x$, $g'(x) = e^x - 1$, negative if x<0 and positive if x>0. The minimum is $g(0) = 0$, which proves the inequality; equality only at 0. The convergence of Euler's method is not proved here.

Worked example. With n=2, Euler gives $y_0 = 1, y_1 = 1.5, y_2 = 2.25$. With n=10, $(1.1)^{10} \approx 2.59374$. We compare with e; we do not confuse an approximate value with the definition of the function.

Watch out

A graph showing a curve above a line does not prove the inequality for all real numbers.

Immediate application. Show that $e^{2x} \geq 1 + 2x$ for every real number x.